

Due: Thursday, September 21st at the beginning of class.

Read Section 0.4 of *Levin* before you start this homework. Then complete the following:

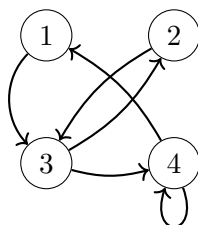
- Chapter 0, Section 4: # 1, 2, 4 - 11 (These do not require explanation, except for # 11.)
- (Optional) Redo the Trials of Xanthius for full points.
- And complete the following:

(1) A **relation** R from a set A to a set B is a subset of $A \times B$. Another definition of **function** from A to B is that it is a relation where each element of A is paired with exactly one element of B . Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$. Determine if each of the following are relations, functions, both, or neither, from A to B . Explain your answer fully.

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| (a) $\{(3, 5), (4, 2), (1, 3)\}$. | (e) $\{(1, 1), (4, 2), (1, 3)\}$. |
| (b) $\{(3, 5), (4, 4), (1, 3), (2, 3)\}$. | (f) $\{(4, 4), (3, 5), (4, 2), (1, 3), (2, 4)\}$. |
| (c) $\{(3, 1), (2, 2), (1, 3)\}$. | (g) $\{(1, 3), (1, 4), (2, 5), (3, 2), (1, 6), (2, 4)\}$. |
| (d) $\{(x, y) \mid x \in A \text{ and } y = x + 2\}$. | (h) $\{(x, y) \in A \times B \mid x < y\}$. |

(2) For each of the parts in the previous problem, sketch a graph and an arrow diagram that represents the ordered pairs. If it is neither a relation nor a function, explain what goes wrong when trying to sketch both the graph and the arrow diagram. (See "Describing Functions" subsection in Section 0.4 for quick examples of these.)

(3) Another way to describe relations from a set X to itself is a variation on the arrow diagram we did in class and in the book. Make a dot on the page anywhere for each element of X and draw an arrow from the dot representing x to y if (x, y) is in the relation. For example, $\{(1, 3), (2, 3), (3, 2), (3, 4), (4, 1), (4, 4)\}$ for $X = \{1, 2, 3, 4\}$ would look like the diagram below. Make a diagram for each of the following relations on $X = \{1, 2, 3, 4\}$.



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| (a) $\{(1, 1), (1, 2), (1, 3), (1, 4)\}$. |
| (b) $\{(1, 2), (2, 3), (3, 4), (4, 1)\}$. |
| (c) $\{(2, 1), (3, 2), (4, 3), (1, 4)\}$. |
| (d) $\{(1, 1), (1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$. |
| (e) $\{(3, 2), (2, 2), (4, 1), (1, 4), (4, 4), (3, 3)\}$. |